ADC Lab Logbook

Part 1 | Section 2: Reactive Components

Section 2: Reactive Components

Before the Lab: Theoretical Preparation

Unlike resistors, capacitors and inductors exhibit frequency-dependent impedance. From the lecture, the impedance of an ideal capacitor is $Z^c = 1/(j\omega C)$, and that of an ideal inductor is $Z^L = j\omega L$, where $\omega = 2\pi f$.

To verify this behavior, a series RL circuit (R = 10 Ω, L = 1 mH) was first analyzed theoretically. The total impedance is given by $Z = j\omega L + R$. The magnitude |Z| and argument arg(Z) were calculated at logarithmically spaced frequencies from 1 Hz to 100 kHz using a spreadsheet:

Frequency (Hz)	|Z| (Ω)	arg(Z) (rad)
1	10.0	0
3.2	10.00002	2.01×10^{-3}
10	10.0001	6.28×10^{-3}
32	10.002	0.0201
100	10.01	0.0627
320	10.2	0.198
1000	11.8	0.561
3200	22.5	1.109
10000	63.6	1.143
32000	201.3	1.521
100000	628.4	1.555

The results were plotted as |Z| vs frequency (log-log scale) and arg(Z) vs frequency. At low frequencies, the inductive reactance is negligible and |Z| ≈ R = 10 Ω. At high frequencies, the inductive impedance dominates, causing |Z| to increase linearly with frequency on the log-log plot. The argument approaches π/2 rad as the inductor becomes the dominant component.

Measurement Method

The impedance measurement technique from Section 1 was applied throughout, using the potential divider relationship $Z = R\cdot(V^{И}/V^{ж})$. The series resistor R was adjusted at each frequency to keep the amplitudes of $V^{И}$ and $V^{ж}$ comparable, minimizing measurement error. Two specific cases were identified:

- When $V^{И}$ is too small (component impedance ≫ R), R should be decreased to improve signal-to-noise ratio.
- When the MATH channel appears blocky or clipped (component impedance ≪ R), R should be increased to reduce loading effects.

Characterizing a Capacitor

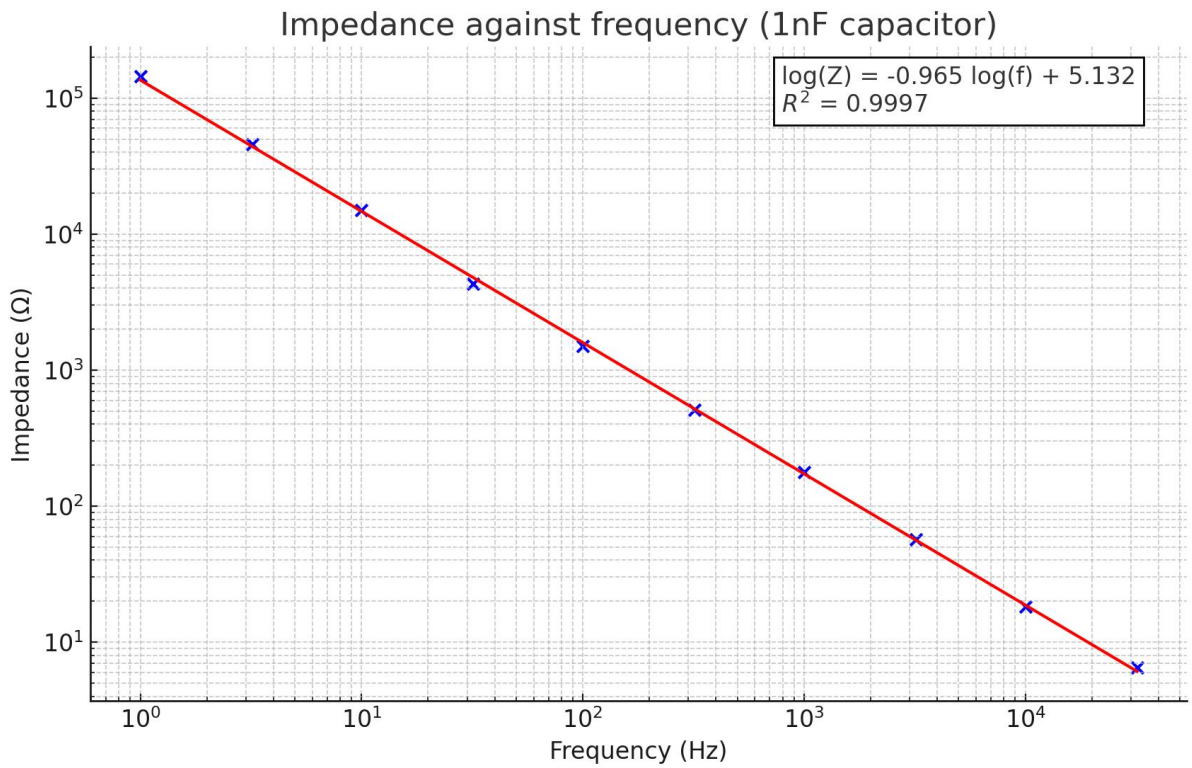
1 µF Capacitor

The impedance of a 1 µF capacitor was measured over the frequency range 1 Hz to 32 kHz. Since capacitive impedance is very large at low frequencies, a starting resistor of R = 100 kΩ was used, adjusted downward as frequency increased.

Frequency (Hz)	R (Ω)	Vz (mV)	Vr (mV)	|Z| (Ω)
1	100k	206	144	144k
3.2	100k	105	230	45.7k
10	100k	37.5	249	15k
32	1k	244	57	4.3k
100	1k	204	133	1.5k
320	1k	106	208	510
1000	1k	40	226	177
3200	100	84	147	57
10000	100	29.6	163	18.2
32000	100	10	154	6.5

All results are based on experimentally measured data, not numerical or circuit simulations.

The data were plotted as log|Z| vs log$f$. A linear fit yielded the relationship $\log(Z) = -0.965 \cdot \log(f) + 5.132$ ($R^2 = 0.9997$). The theoretical gradient for an ideal capacitor is −1, and the measured value of −0.965 is in close agreement, confirming the expected capacitive behavior. Logarithmic scaling was employed to clearly resolve the wide dynamic range of impedance values across the frequency sweep.



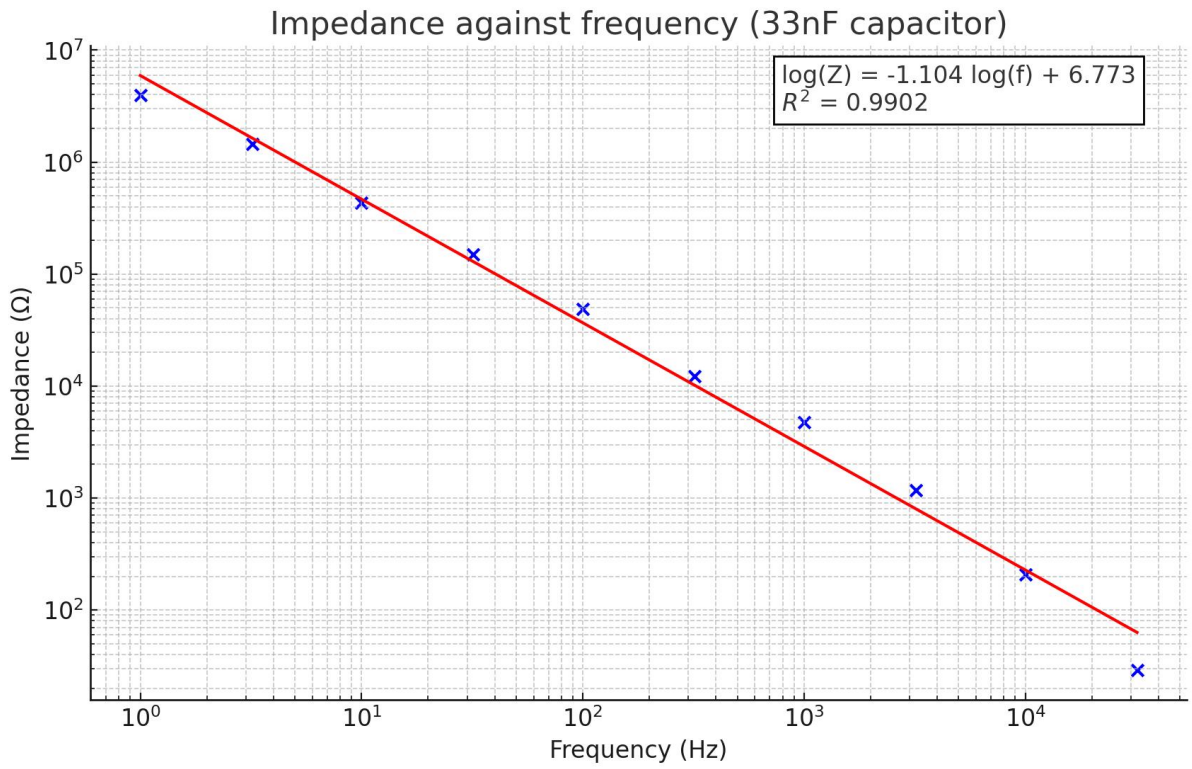
33 nF Capacitor

A second capacitor of 33 nF was characterized over the same frequency range. A 1 MΩ starting resistor was used at low frequencies due to the much higher impedance of this smaller capacitance.

Frequency (Hz)	R (Ω)	Vz (mV)	Vr (mV)	|Z| (Ω)
1	1M	224	56.6	3958k
3.2	1M	194	133.1	1458k
10	100k	242	56	432k
32	100k	206	139	149k
100	100k	110	226	48.7k
320	1k	248	20.3	12.2k
1000	1k	246	51.9	4.74k
3200	1k	233	200	1.17k
10000	10	248	12	206.7
32000	1	248	7.9	28.9

All results are based on experimentally measured data, not numerical or circuit simulations.

The log-log linear fit gave $\log(Z) = -1.104 \cdot \log(f) + 6.773$ ($R^2 = 0.9902$), again close to the theoretical gradient of −1.



Characterizing an Inductor

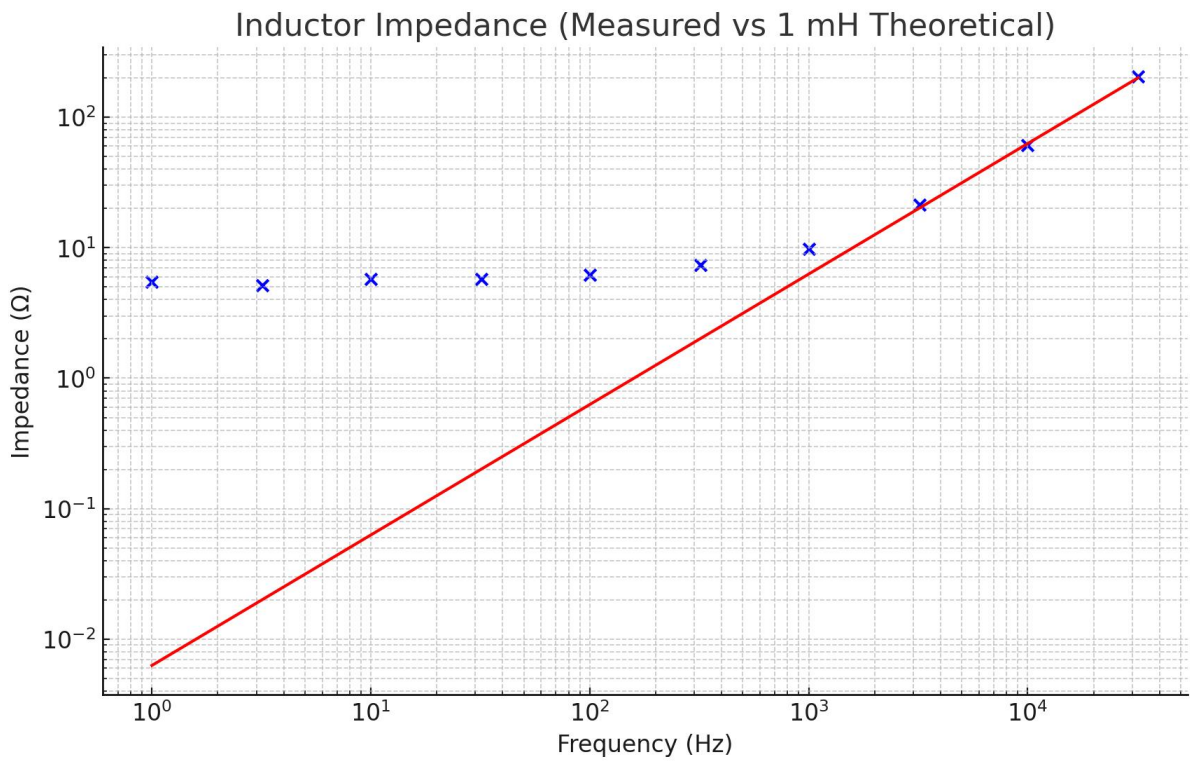
1 mH Inductor

The 1 µF capacitor was replaced by a 1 mH inductor. Since inductive impedance is proportional to frequency, it is very small at low frequencies. Accordingly, a starting resistor of R = 1 Ω was used and increased progressively with frequency.

Frequency (Hz)	R (Ω)	Vz (mV)	Vr (mV)	|Z| (Ω)
1	1	37.2	6.8	5.47
3.2	1	38	6.2	5.13
10	1	38	6.6	5.76
100	1	37.2	6	6.2
1000	10	38.4	39	9.85
3200	10	73	34	21.47
10000	100	88	144	61.1
32000	1k	44.8	219.2	204.38

All results are based on experimentally measured data, not numerical or circuit simulations.

The measured data were plotted alongside a theoretical line derived from $Z = j\omega L$. At low frequencies, the measured impedance plateaus rather than continuing to decrease, revealing the presence of the inductor's parasitic (series) resistance. The theoretical line assumes an ideal inductor and thus continues to decrease, whereas the measurement converges to a horizontal asymptote equal to the parasitic resistance.

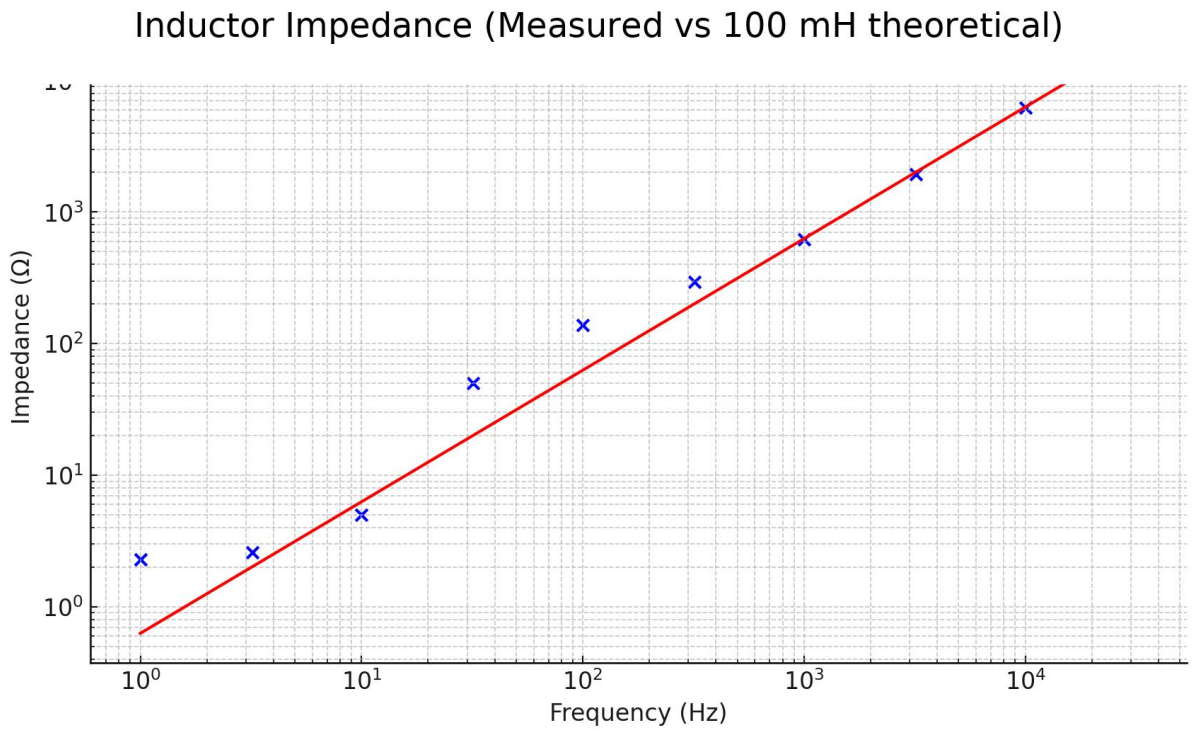


100 mH Inductor

A 100 mH inductor was also characterized using the same method over the range 1 Hz to 32 kHz.

Frequency (Hz)	R (Ω)	Vz (mV)	Vr (mV)	|Z| (Ω)
1	1	200	85.6	2.3
3.2	1	208	79.3	2.6
10	1	204	41.1	5
32	10	350	70	50
100	10	392	28.5	138
320	100	352	120	293
1000	100	452	73.2	618
3200	1k	416	215	1935
10000	10k	264	425	6212
32000	10k	450	216	20833

All results are based on experimentally measured data, not numerical or circuit simulations.



Discussion: Sources of Measurement Discrepancy

Two principal sources of deviation between measured and theoretical impedance values were identified:

1. Component tolerances. Real passive components are manufactured with finite precision. For example, a nominal 100 Ω resistor may have an actual value in the range 95–105 Ω. This was directly observed in Section 1, where the nominally 10 kΩ resistor measured 9.74 kΩ, representing a 2.6% deviation.

2. Parasitic resistance. Neither capacitors nor inductors are ideal reactive elements in practice. A more accurate model includes a parasitic series resistance R , such that:

$$Z^c = 1/(j\omega C) + R \qquad Z^L = j\omega L + R$$

For a capacitor, as frequency increases, 1/(jωC) tends to zero, and the impedance approaches R rather than zero. For an inductor, as frequency decreases toward zero, the impedance approaches R rather than zero — consistent with the horizontal asymptote observed in the 1 mH inductor plot.

Challenge: Determining Unknown Component Values

The impedance method was applied to determine the capacitance and inductance of two unknown capacitors and two unknown inductors. Measurements were taken at five frequencies between 1 kHz and 10 kHz, and the component values were extracted using:

$$C = 1/(2\pi f \cdot |Z|) \qquad L = |Z|/(2\pi f)$$

The calculated values were subsequently verified using an RLC bridge for direct measurement.

Capacitors

Freq (Hz)	Cap 1: R (Ω)	Vz	Vr	|Z| (Ω)	C (nF)	Cap 2: R (Ω)	Vz	Vr	|Z| (Ω)	C (nF)
1k	1k	238	68	3500	45.5	10k	152	192	7916.7	20.1
2.5k	1k	200	136	1470.6	43.3	10k	76	232	3275.9	19.4
5k	1k	142	190	747.4	42.6	1k	204	132	1545.5	20.6
7.5k	1k	106	210	504.8	42.1	1k	173	164	1054.9	20.1
10k	1k	84	220	381.8	41.7	1k	148	185	800	19.9
Measured (RLC)					48 nF					21.9 nF

Inductors

Freq (Hz)	Ind 1: R (Ω)	Vz	Vr	|Z| (Ω)	L (mH)	Ind 2: R (Ω)	Vz	Vr	|Z| (Ω)	L (mH)
1k	100	34.8	152	22.9	3.64	100	146	104	140.4	22.3
2.5k	100	76	146	52.1	3.32	100	214	66	324.2	20.6
5k	100	132	130	101.5	3.23	1k	129	190	678.9	21.6
7.5k	100	168	111.5	150.7	3.20	1k	165	166	994	21.1
10k	100	194	96	202.1	3.22	1k	190	144	1319.4	21

Measured (RLC)					3.28 mH					21.6 mH

All results are based on experimentally measured data, not numerical or circuit simulations.

A systematic discrepancy was observed between the impedance-derived values and those measured by the RLC bridge. This is attributed to the parasitic resistance present in both components: the impedance method measures the total complex impedance |Z|, which includes the contribution of R and therefore overestimates the reactive component. By contrast, the RLC bridge employs an internal algorithm that separates the reactive and resistive parts, enabling a more accurate determination of the pure capacitance or inductance.